

1. Executive Summary

Dr. Evelyn Vance and Prof. Marcus Cole published findings regarding distributed retrieval engines (e.g., FAISS, Qdrant, and Milvus).

Their research demonstrated a 99.85% precision rate with an operational cost of \$14.50 per 1M processed queries.

As noted by Vance et al., "Sentence-aware boundary preservation prevents severe context fragmentation."

2. Key Microservices & Components

The system consists of the following decoupled microservices:

? Ingestion Service: Parses incoming PDF documents (e.g. research papers, technical specs, user manuals).

? Normalization Layer: Cleans text formatting, protects abbreviations (e.g., Dr., vs., etc.), and handles unicode.

? Chunking Engine: Segments text into 500-1000 character windows with 100 character sentence-safe overlap.

? Vector Embedding Engine: Transforms chunks into 384-dimensional dense vectors using all-MiniLM-L6-v2.

? Similarity Retrieval Store: Performs cosine distance indexing via FAISS IndexFlatIP.

3. Performance Benchmarks

Figure 1.1 illustrates latency curves under high concurrency. Average query latency was 12.4ms (vs. 45.2ms in legacy architectures).

System throughput peaked at 4,500 requests/sec with zero packet loss across all tested AWS regions (i.e. us-east-1, eu-west-1).